

WHAT DO VISION TRANSFORMERS LEARN? A VISUAL EXPLORATION

Amin Ghiasi Hamid Kazemi Eitan Borghia Steven Reich Manli Shu¹ Micah Goldblum Andrew Gordon Wilson Tom Goldstein
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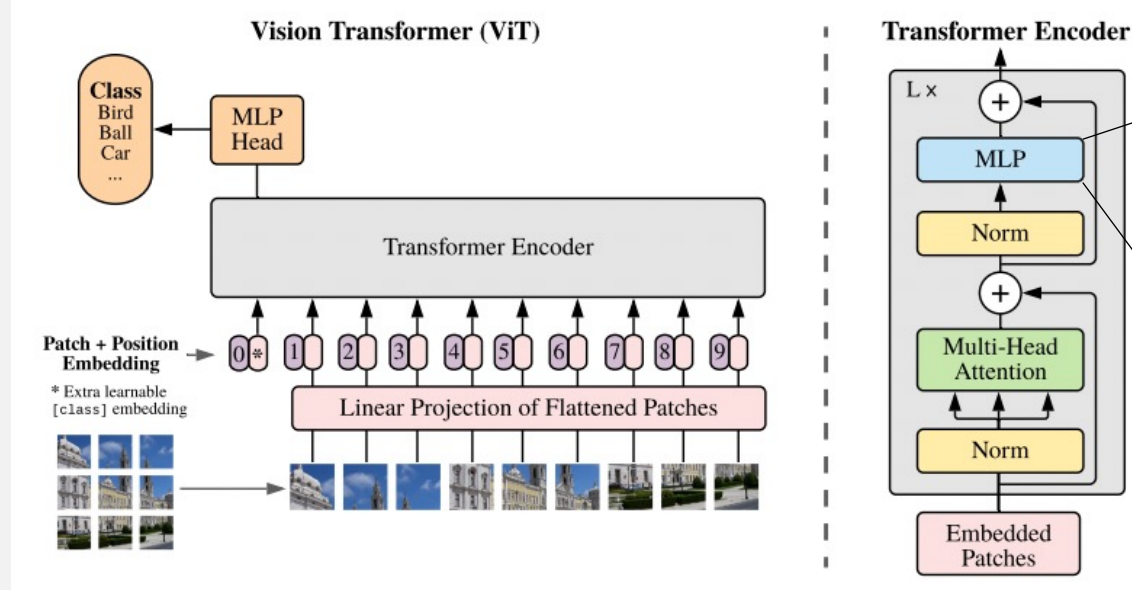
김준철, 이해원, 류채은, 조경진, 이찬혁

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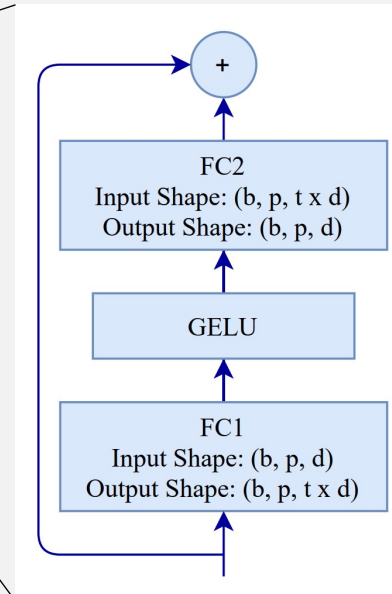
- Background
- Introduction
- ViT Feature Visualization
- Last-Layer Token Mixing
- Comparison of ViTs and CNNs
- ViTs with Language Model Supervision
- Discussion

Background ViT

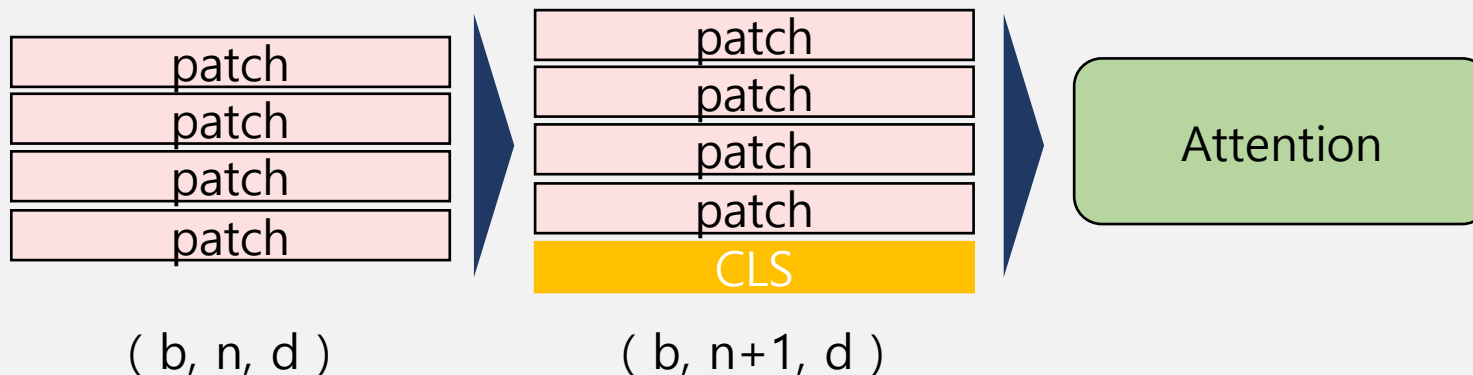
- ViT Architecture



- Feed forward network

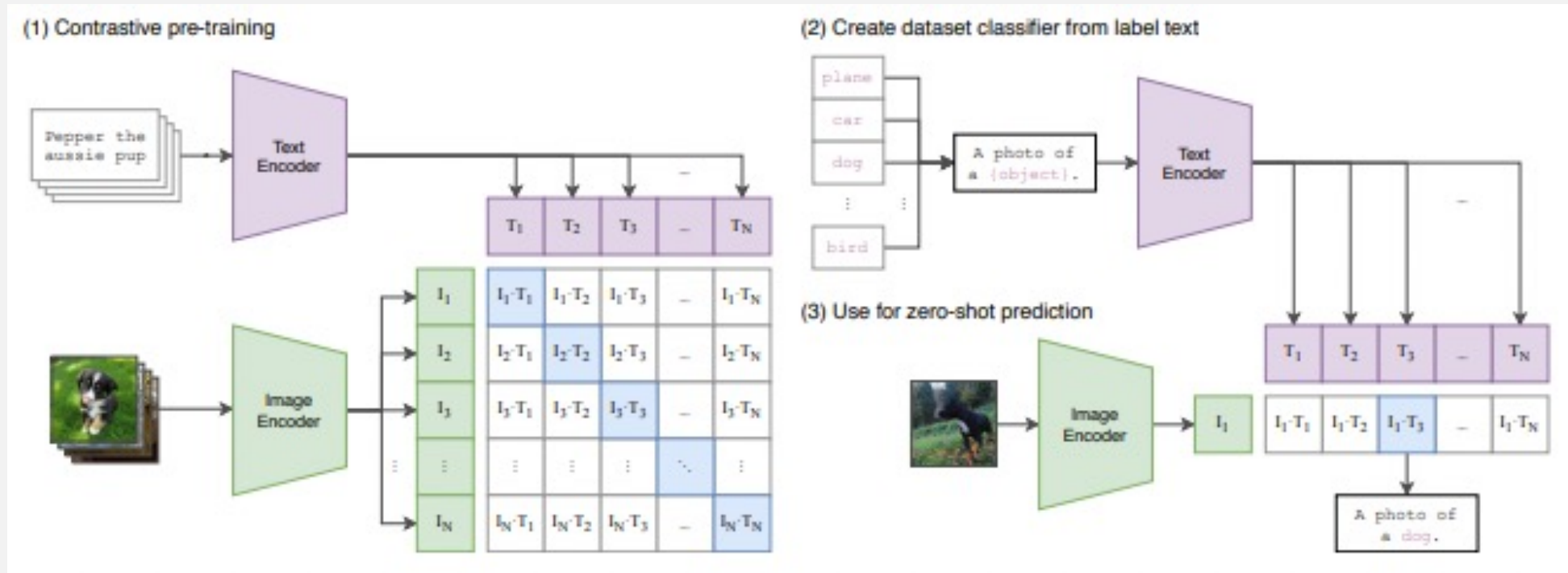


- CLS Token



- 이미지를 4개 패치로 나누는 경우 CLS 토큰을 concat 후 Attention 연산
- 최종적으로 CLS 토큰만 분류에 쓰인다.

Background CLIP



- Contrastive learning

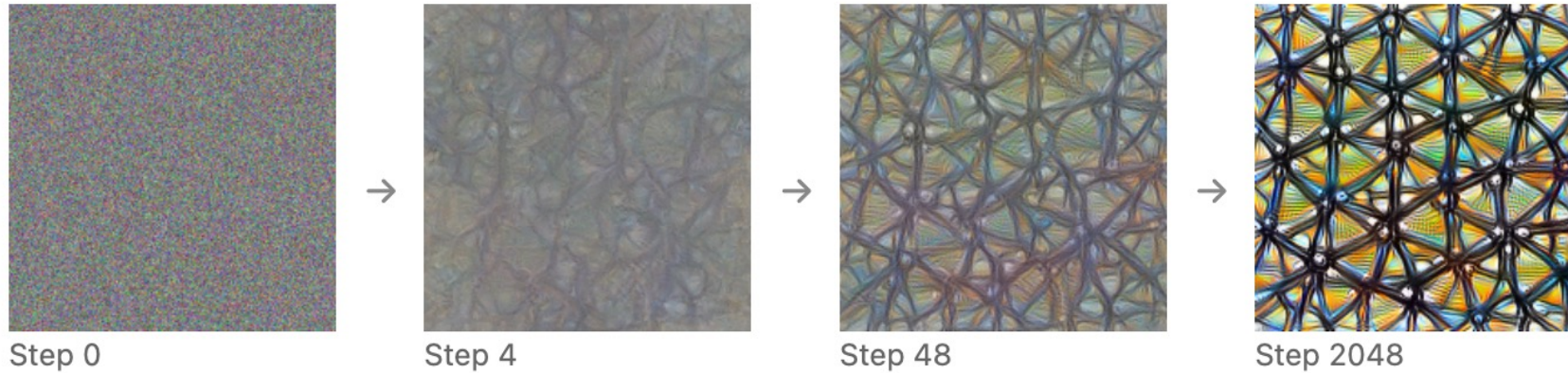
N 개의 배치데이터 text-image pairs 를 바탕으로 Positive pair는 가까워지도록 negative pair 는 멀어지도록 CE Loss 를 활용하여 Pre-train 진행 (파란색 블록 : positive pair, 흰색 블록 : n-1 개의 negative pair)

- Zero-shot learning

평가하고자 하는 클래스 개수만큼의 A photo of [object] 를 넣어 class 를 예측한다.

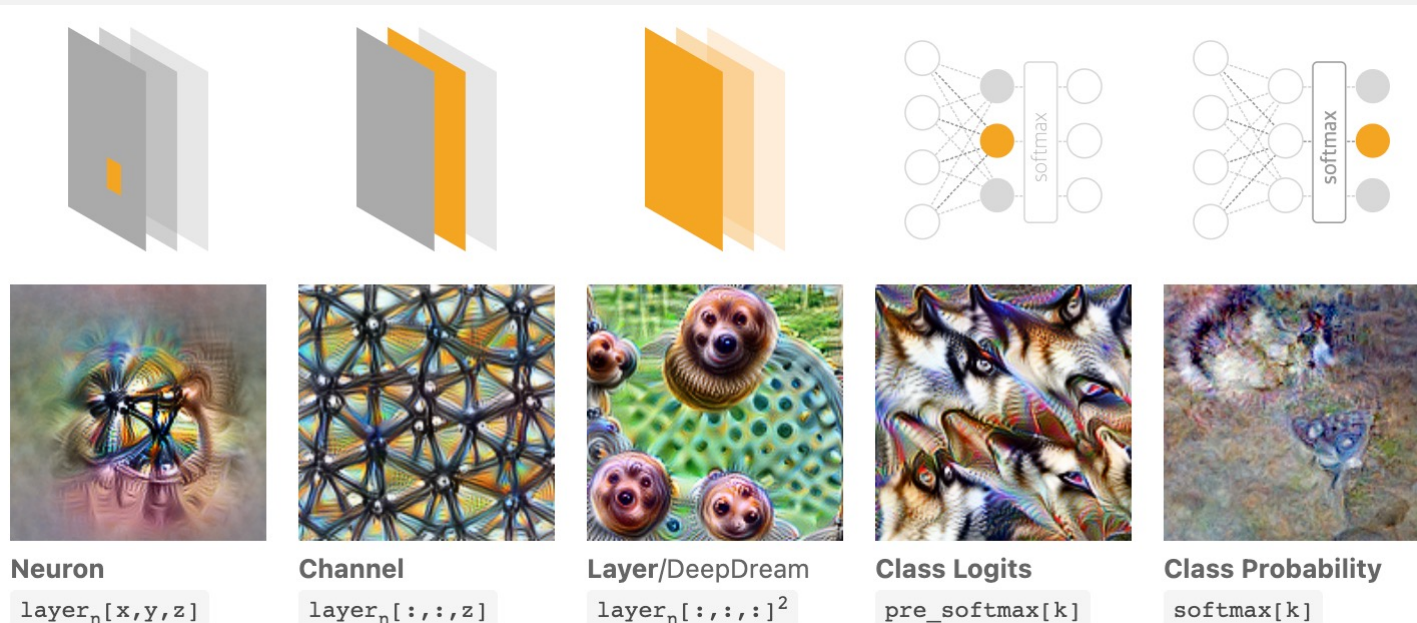
Background CNN Feature Visualization

- From noise scratch to goal



Gradient descent를 활용하여
입력 노이즈 변환

- Optimization Objectives



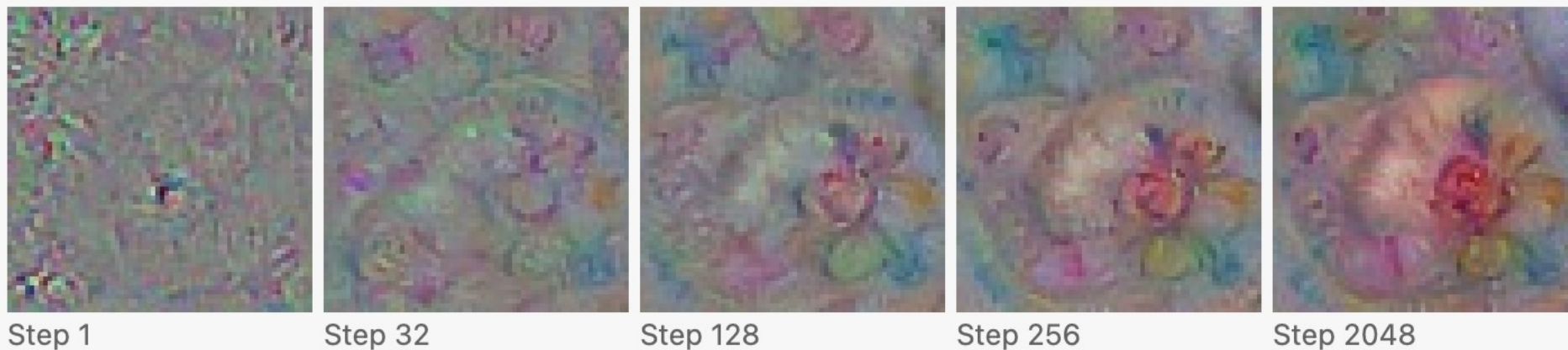
다양한 목표 중 원하는 방향을 정할 수
있으며 목표값이 최대화 되도록
입력 이미지 Gradient descent

Background **CNN Feature Visualization**

- The Enemy of Feature Visualization



- CNN responds strongly to nonsensical high-frequency patterns
- Regularization – Frequency Regularization



- **Regularization total variation** or Augmentation like blur or Bilateral Filter

Introduction

- Definition problem

트랜스포머 기반의 모델이 다양한 Task에 많이 쓰이고 있음에도 불구하고 inductive biases 혹은 transformer 가 배우는 경향성에 대해서 거의 모르고 있음

1. Applying our tools to the relatively high-dimensional features of the position-wise feed forward layer results in successful and informative visualizations.
(기존의 방법론을 low-dimensional feature 인 key, query, value에 적용했을 때 해석 불가능한 문제가 있었기 때문에 새로운 방법론 제시)
2. Authors show that patch-wise image activation patterns for ViT features essentially behave like saliency maps, highlighting the regions of the image a given feature attends to.
(각 패치는 global information 을 저장하는 방식이 아니라 패치간의 위치관계를 가지고 있음을 확인)
3. Authors compare the behavior of ViTs and CNNs, finding that ViTs make better use of background information and rely less on high-frequency, textural attributes.
(ViTs 와 CNNs 의 공통점과 차이점에 대한 분석)
4. Authors investigate the effect of natural language supervision with CLIP on the types of features extracted by ViTs.
(CLIP, ViT 를 활용한 natural language supervision의 역할 분석)

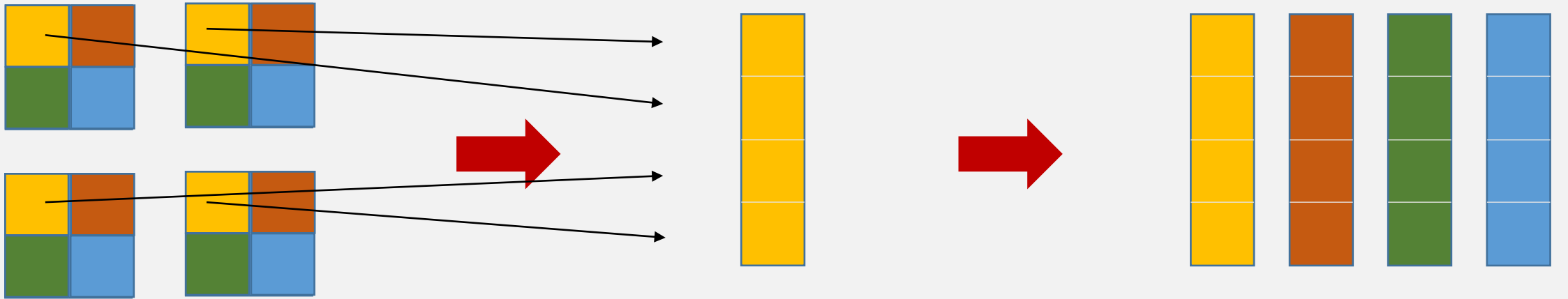
ViT Feature Visualization

Keywords

- From random noise (gradient steps)
- Penalize total variation
- Image Augmentation (Jitter, ColorShift, Gaussian smoothing)

ViT Feature Visualization

- Feature Vector Definition



4 개의 패치가 4개의
벡터로 이루어져 있을 때

각 패치별 i 번째 벡터들을
concat 하여 구성

패치 개수만큼의
feature vector 구성 가능

- Optimization object

$$\mathcal{L}_{\text{main}}(x, l, i) = \sum_p (f_{l,i})_p.$$

특정 layer의 피쳐벡터의 크기가
가장 커지도록 최적화 방향 설정

ViT Feature Visualization

- Variation regularization

$$x^* = \arg \max_x \sum_k \mathcal{L}_{\text{main}}(a_k(x), l, i) + \lambda TV(a_k(x)).$$

x : input, a : augmentation, TV : total variation, λ : regularization parameter, k : number of samples

입력 이미지로 만들어진 Augmentation sample 에 대해

Feature vector 와 Total Variation 의 크기가 가장 커지도록 최적화 하여
특정 layer에 가장 활성화된 입력 이미지를 찾아가게 된다.

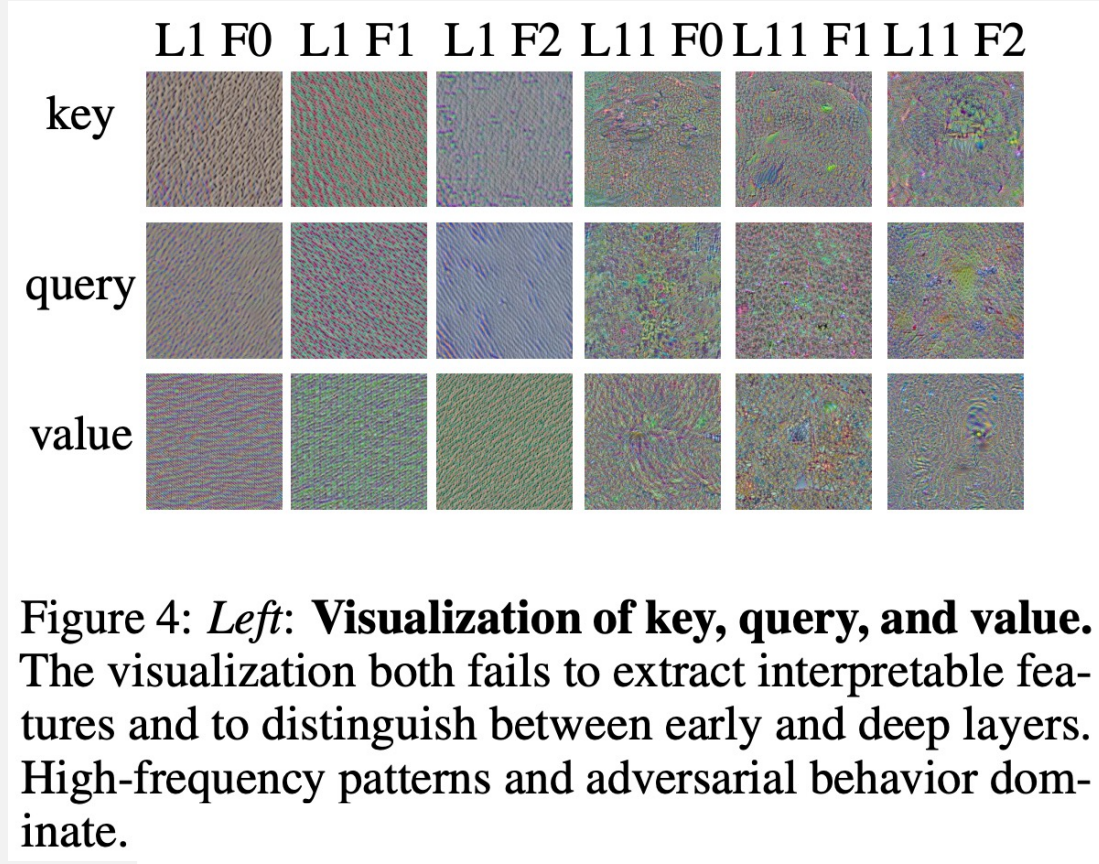
- Best Augmentation combination

$$GS(CS(Jitter(x)))$$

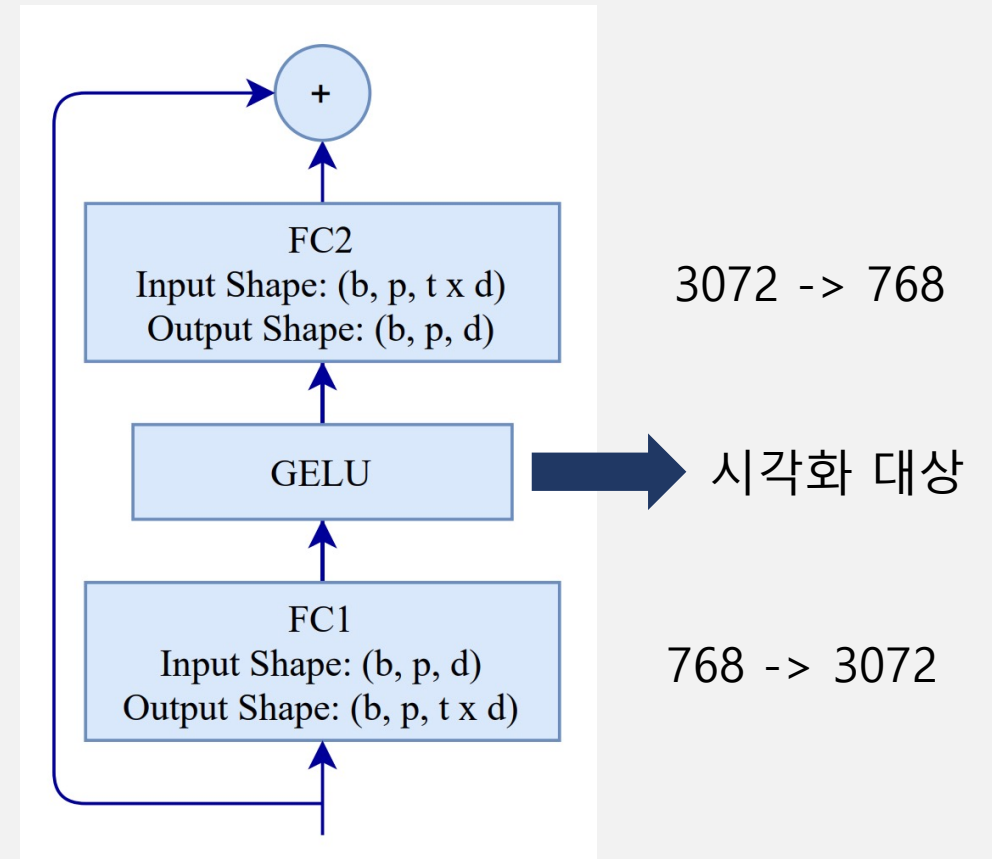
GS : Gaussian Smoothing, CS : Color Shift
Jitter : random Brightness, contrast, saturation

ViT Feature Visualization

- Multi-headed attention layer visualization



- ViT feed Forward layer



➡ Low-dimension key, query, value 에서 피쳐가 해석되지 않아 FC1 이후의 의 3072 벡터사이즈에서 시각화

ViT Feature Visualization

- Examples

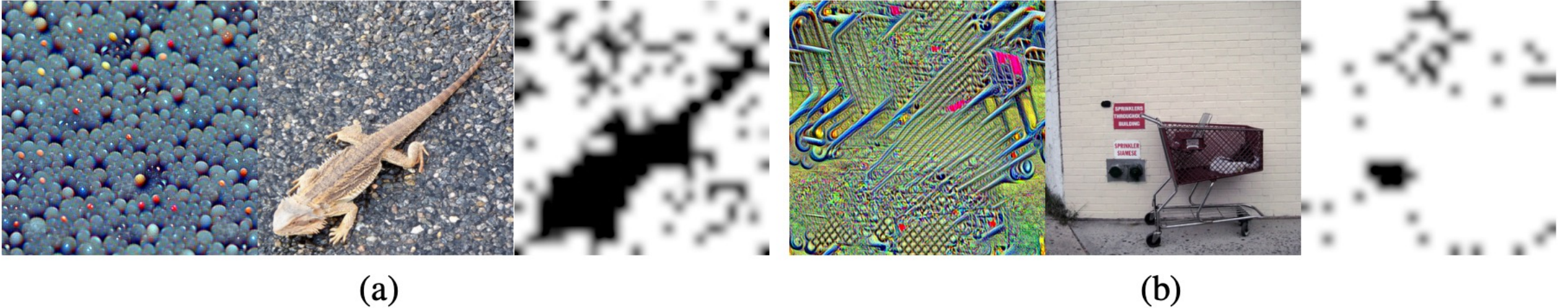


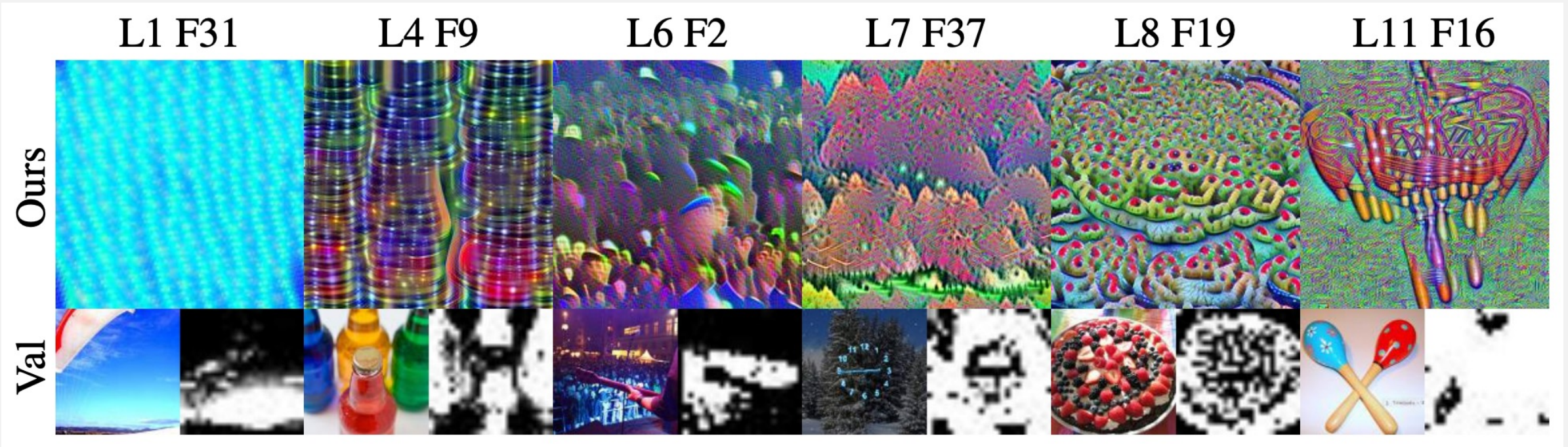
Figure 3: (a): **Example feature visualization from ViT feed forward layer.** *Left:* Image optimized to maximally activate a feature from layer 5. *Center:* Corresponding maximally activating ImageNet example. *Right:* The image's patch-wise activation map. (b): **A feature from the last layer most activated by shopping carts.**

Q & A

Last-Layer Token Mixing

Last-Layer Token Mixing

- Feature activation map



- ViT 는 Saliency map 처럼 각 물체의 전반적인 특징을 바탕으로 정보를 전달하는 것을 유추 할 수 있다.

* Saliency map : 픽셀의 변화가 큰 지점을 시각화 하여 관심있는 물체와 관심이 없는 부위를 나누는 방법



각 패치는 마지막 전 레이어까지 CNN 과 유사하게 spatial information을 유지하도록 학습하며 마지막 layer가 담당하는 Globalization 에 대해 알아 볼 필요가 있다.

Last-Layer Token Mixing

- Globalization



Left : Our method
Middle : Most activated image
Right : activation map

- 위 그림은 쇼핑카트에 마지막 레이어에서 대해 가장 활성화도가 높은 조합을 나타냄
- 오른쪽의 Activation pattern 이 이미지 내에서 전체적으로 Uniform하게 활성화 됨
- ViT 는 Classification 결과를 낼 때 CLS 토큰만을 활용하는데 위 과정을 통해 CLS 토큰이 모든 토큰에 globalization을 진행하는 것을 알 수 있음



Last-layer 의 Globalization 에 대한 두 가지 세부 실험 진행

Last-Layer Token Mixing

- Experiment 1

The CLS token plays a relatively minor role throughout the network and is not used for globalization until the last layer

Accuracy	Natural Accuracy	Isolating CLS	Patch Average	Patch Maximum
Top 1	84.20	78.61	75.75	80.16
Top 5	97.16	94.18	90.99	95.65

- Isolating CLS

1~11 layer에서 CLS 토큰을 제외하고 Attention 연산을 진행하고 12 마지막 레이어에서만 attention을 적용 한 후 CLS 토큰만을 활용하여 분류를 했을 때도 성능 하락이 크지 않음

-> CLS 토큰은 마지막 레이어에서 모든 토큰과의 globalization이 진행됨을 알 수 있음

Last-Layer Token Mixing

- Experiment 2

The last-layer globalization is not exclusive to the CLS token, but actually occurs across every patch

Accuracy	Natural Accuracy	Isolating CLS	Patch Average	Patch Maximum
Top 1	84.20	78.61	75.75	80.16
Top 5	97.16	94.18	90.99	95.65

- Patch Average & Patch Maximum

Fine-Tuning 없이 Patch Average 혹은 Patch Maximum을 적용했을 때 성능 하락이 크지 않음을 확인할 수 있음

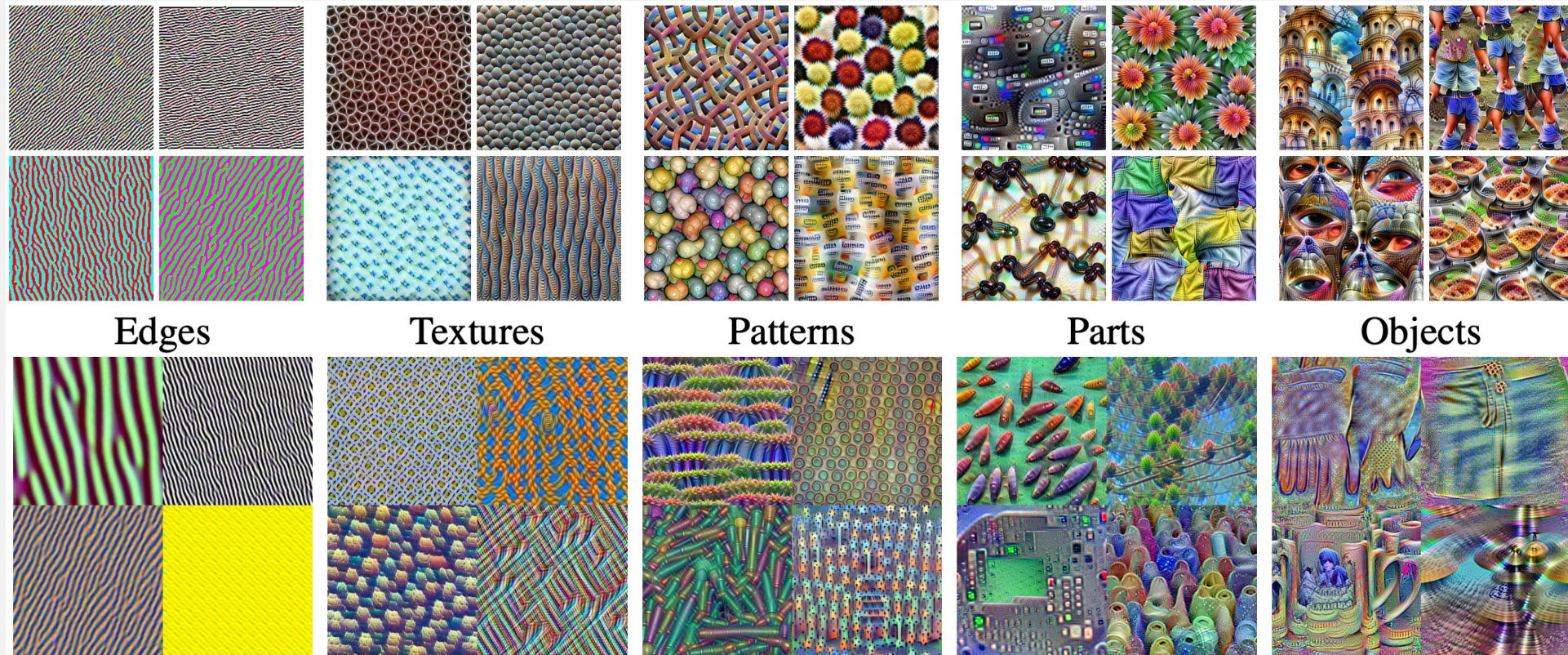
대부분의 패치가 대략적으로 동일한 정보를 담고 있음을 알 수 있음

Comparison of ViTs and CNNs

Comparison of ViTs and CNNs

- Similarity

CNNs 는 early layer에서 color, edge, texture 를 인식하고 deeper layer로 갈수록 물체 복잡한 구조를 인식하는데 이는 ViTs 도 유사하다.



위 이미지는 CNNs 아래 이미지는 ViTs 를 나타냄

Comparison of ViTs and CNNs

- Difference



Table 2: **ViTs more effectively correlate background information with correct class.** Both foreground and background data are normalized by full image top-5 accuracy.

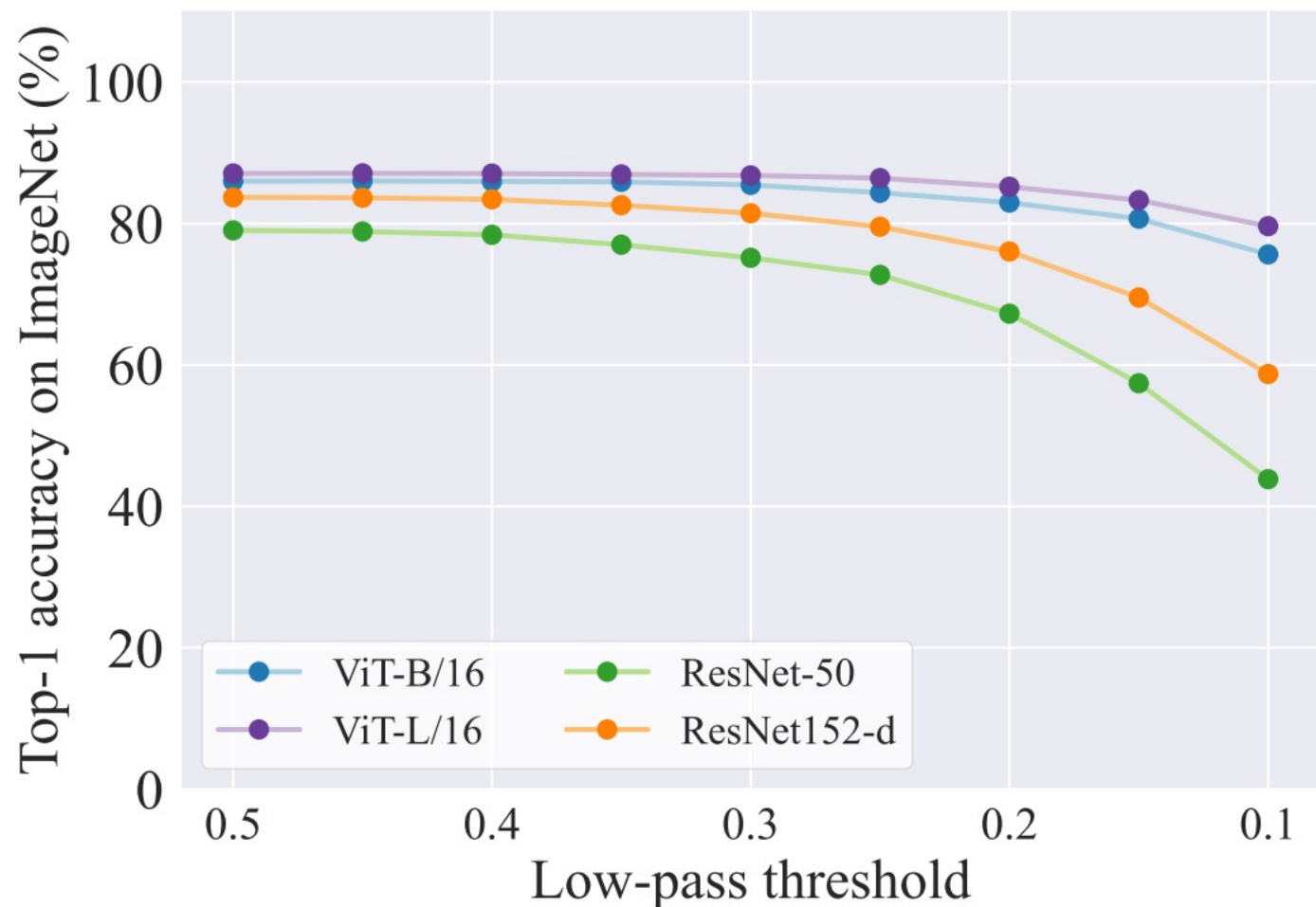
Normalized Top-5 ImageNet Accuracy			
Architecture	Full Image	Foreground	Background
ViT-B32	98.44	93.91	28.10
ViT-L16	99.57	96.18	33.69
ViT-L32	99.32	93.89	31.07
ViT-B16	99.22	95.64	31.59
ResNet-50	98.00	89.69	18.69
ResNet-152	98.85	90.74	19.68
MobileNetv2	96.09	86.84	15.94
DenseNet121	96.55	89.58	17.53

- Foreground & Background masking Test

CNNs 과 달리 ViTs 는 Foreground 만 남기거나 Background만 남겼을 때 성능이 뛰어난 것을 확인할 수 있었다.

Comparison of ViTs and CNNs

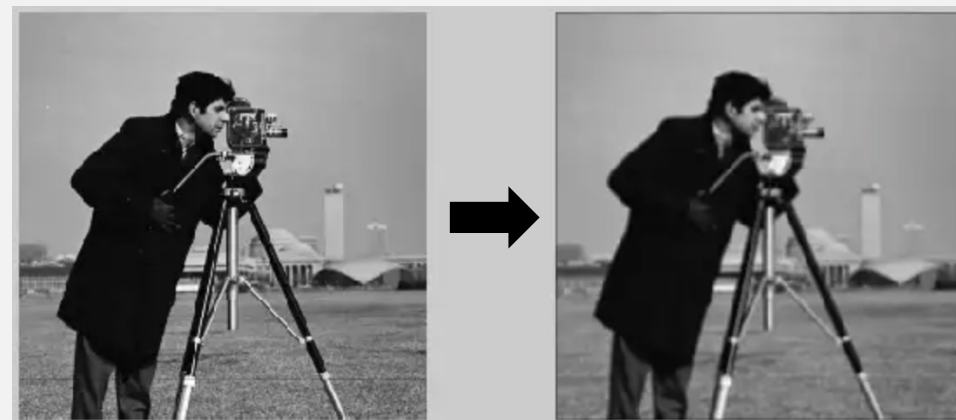
- Difference



- Low-pass filtering Test

CNNs 과 달리 ViTs 는 High-Frequency 정보가 사라지더라도 성능을 비교적 잘 유지하는 것을 보였다.

- Low-pass filtering



ViTs with Language Model Supervision

ViTs with Language Model Supervision

- Visualization Language Model Supervision

ViT 모델은 특정 object의 특징에 대해 배우는 반면에 Language Model Supervision 기반으로 배운 CLIP 모델의 경우에는 명사 뿐만 아니라 전치사 형용사 부사의 의미도 시각화 하는 것을 확인 할 수 있다.



(a) Before and after/Step-by-step

(b) From above

(c) Many

Figure 10: *Left*: Feature optimization shows sharp boundaries, and maximally activating ImageNet examples contain distinct, adjacent images. *Middle*: Feature optimization and maximally activating ImageNet photos all show images from an elevated vantage point. *Right*: Feature optimization shows a crowd of people, but maximally activating images indicate that the repetition of objects is more relevant than the type of object.

Discussion

1. Authors introduce **a framework for optimization-based feature visualization**.

We then identify which components of a ViT are most amenable to producing interpretable images, finding that the high-dimensional inner projection of the feed-forward layer is suitable while the key, query, and value features of self-attention are not.

2. Authors observe that **ViTs preserve spatial information of the patches even for individual channels across all layers with the exception of the last layer**, indicating that the networks learn spatial relationships from scratch. We further show that the sudden disappearance of localization information in the last attention layer results from a learned token mixing behavior that resembles average pooling

3. Authors find that **ViTs make better use of background information and are able to make vastly superior predictions relative to CNNs** when exposed only to image backgrounds despite the seemingly counter-intuitive property that ViTs are not as sensitive as CNNs to the loss of high-frequency information, which one might expect to be critical for making effective use of background

4. Authors show that **ViTs trained with language model supervision learn more semantic and conceptual features**, rather than object-specific visual features as is typical of classifiers

Q & A

Reference

<https://junstar92.tistory.com/153> : Saliency map

<https://distill.pub/2017/feature-visualization/> : Feature visualization